What is Convoy Effect | FCFS CPU Scheduling Algorithm

Introduction to Process Scheduling

1. Process Scheduling

It is the method used by the Operating System to decide **which process will run on CPU next**.

- It is the base of multiprogramming OS.
- OS keeps switching CPU between many processes so the computer works faster.
- Many processes stay in memory at the same time. If one process is waiting or its time slice is over, OS gives CPU to another process.

Example:
Imagine 5 students (processes) want to use a single pen (CPU).
Each student gets the pen one by one → this is scheduling.

2. CPU Scheduler

This selects which process from the **ready queue** will run on CPU next.

- When CPU becomes free, scheduler picks one process.
- This is done by **Short Term Scheduler (STS)**.

Example:
CPU is like a bus seat. When one person gets off, the conductor chooses next passenger in line.

3. Non-Preemptive Scheduling

Once a process gets the CPU, **it will keep using it until it finishes or waits**.
No one can take CPU from it forcefully.

- Long processes can delay small processes → **starvation** may occur.
- CPU utilization becomes low in some cases.

Example:
If a teacher gives the pen to one student, he keeps it until he finishes writing.
Even if other students need only 10 seconds, they must wait.

4. Preemptive Scheduling

Here OS can **take CPU back from a process** after time quantum or due to wait/termination, and give it to someone else.

- Less starvation than non-preemptive.
- CPU utilization becomes better.

Example:
Teacher gives every student pen only for 2 minutes.
After 2 minutes, even if unfinished, pen goes to the next student.
Everyone gets a fair chance.

5. Goals of CPU Scheduling

CPU scheduling aims to improve system performance:

a. Maximum CPU use
b. Minimum Turnaround time (TAT)
c. Minimum Waiting time
d. Minimum Response time
e. Maximum Throughput

Simple Understanding:
System should be fast, responsive, and complete more tasks in less time.

6. Throughput

Number of processes completed per unit time.

📌 Example:
If in 1 hour CPU completes 20 processes → throughput = 20/hr.

7. Arrival Time (AT)

Time when a process enters the ready queue.

📌 Example:
P1 arrives at 0 sec, P2 at 2 sec → P1 AT=0, P2 AT=2.

8. Burst Time (BT)

Execution time required by a process.

📌 Example:
P1 needs 5 sec CPU time → BT = 5

9. Turnaround Time (TAT)

Time from entering ready queue to completing process.

📌 Formula:
TAT = Completion Time - Arrival Time

📌 Example:
Arrival=0, Completion=10 → TAT = 10

10. Waiting Time (WT)

Time a process spends waiting in ready queue.

📌 Formula:
WT = TAT - BT

📌 Example:
TAT=10, BT=6 → WT = 4 sec waiting.

11. Response Time

Time between entering ready queue and getting CPU first time.

📌 Example:
Process arrived at 0, got CPU at 4 → Response Time = 4.

12. Completion Time (CT)

Time when process finishes execution.

📌 Example:
If process stops at time 15 → CT = 15.

13. FCFS (First Come First Serve)

The process that arrives first gets CPU first.
But if the first process has large Burst Time, others wait a long time → **convoy effect**.

Example with times:

Process	Arrival	BT
P1	0	10
P2	1	2
P3	2	1

P2 and P3 wait because P1 is long → **convoy effect**.

Convoy Effect

Many small processes are forced to wait behind a long process.

This cause poor resource management.
This wastes CPU time and slows system.

📌 Real Example:
A slow truck (long process) is in front and many fast cars (short processes) are stuck behind → all become slow.